

JANE SIMKO

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Postdoctoral Fellowships

Sep 2022 - Sep 2025 Makinson Lab, Columbia University, Department of Neurology
BRAIN Initiative Individual Postdoctoral Fellowship (F32)
New York, NY U.S.A

Mar 2021 - Aug 2022 Makinson Lab, Columbia University, Department of Neurology
Swiss National Science Foundation Early Postdoc.Mobility Fellowship
New York, NY U.S.A

Education

Jan 2016 - Jan 2021 École Polytechnique Fédérale de Lausanne
PhD in Neuroscience
Supervisor: Prof. Henry Markram, Co-Supervisor: Prof. Sean Hill
Lausanne, Vaud, Switzerland

Sep 2013 - Jan 2015 University of Lausanne
Masters in Medical Biology: Neuroscience
Lausanne, Vaud, Switzerland

Aug 2007 - May 2011 University of California, Berkeley
B.A. Biological Sciences
Berkeley, California, United States

Academic Research Projects

Jan 2021 - present Chris Makinson Lab, Department of Neurology, Columbia University
Postdoc under Prof. Chris Makinson

- Executing behavioral paradigms to probe sensory perception and decision-making in rodents, integrating real-time electrophysiological and behavioral readouts
- Building CNN-based classification pipelines for online EEG/EMG decoding

Jan 2016 - Jan 2021 Henry Markram Lab, Laboratory of Neural Microcircuitry, EPFL
Sean Hill Lab, Laboratory for the Neural Basis of Brain States, EPFL
PhD Thesis under Prof. Henry Markram and Prof. Sean Hill

- Utilizing multiple patch clamping strategies for elucidating microcircuitry in the rodent somatosensory thalamus
- Creating comprehensive thalamocortical models and simulations in collaboration with the Blue Brain Project

Jun 2019 - Oct 2019 John Huguenard Lab, Department of Neurology, Stanford
EPFL-Stanford Exchange Program under Prof. John Huguenard

- Performing stereotaxic GRIN lens implantations in mice
- Building and troubleshooting custom 2p galvo and resonant scanning microscopes

Mar 2014 - Jan 2015 Paola Bezzi Lab, UNIL Department of Fundamental Neurosciences
Master's Thesis under Francesco Petrelli, PhD

- Investigating the role of astrocytes in dopamine regulation in the prefrontal cortex of mice
- Performing viral injections of channelrhodopsin constructs in mice, molecular biology
- Running *in vivo* optogenetic laser stimulation and microdialysis protocols

Oct 2013 - Dec 2013 Richard Benton Lab, UNIL Center for Integrative Genomics
Dr. Pavan Ramdya, EPFL Laboratory of Intelligent Systems
First Semester Masters Project under Pavan Ramdya, PhD and Prof. Richard Benton

- Mapping the mechanosensory neural circuits of *Drosophila* to inspire the construction of sensitive prosthetics and advanced walking robots
- Programming MATLAB code for *Drosophila* image manipulation

Voluntary Research Projects

Dec 2012 - Apr 2013 Michel Maharbiz Lab, UC Berkeley Dept. of Neuro-EECS
Junior Specialist under Prof. Michel Maharbiz and PhD candidate Peter Ledochowitsch

- Investigating the role of corticostriatal plasticity in abstract skill learning using ECoG microarrays
- Recording microarray data with MATLAB

Jan 2011 - May 2011 Yang Dan Lab, UC Berkeley, Helen Wills Neuroscience Institute
Research Assistant under PhD candidate, Julia Cox

- Examining the effect of learning on visual processing in the visual cortex of mice
- Performing surgical implantations of recording electrodes in the brain and muscle of laboratory animals

Mentorship

Tamina Keira Ramirez – PhD student, Neuroscience, Columbia University
2021–present

Ellie Sung – Undergraduate, Barnard College

2024–present

Sally Leung – M.A. student, Columbia University (now PhD student in Kinesiology at Columbia University)

Fall 2023

Arjun Jadhav – Undergraduate, UC Berkeley

Summer 2023

Eda Akca – High school student (now undergraduate at Northwestern University)

Summers 2022 & 2023

Cece Peters – High school student (now undergraduate at Colgate University)

Summers 2022 & 2024

Teaching Activities

Feb 2020 - Jun 2020 Graduate Student Instructor for Genetics and Bioinformatics

- Creating exercise sets and exam questions.

- Guiding discussions and exercises

Feb 2019 - Jun 2019 Graduate Student Instructor for Physiology Lab Practical

- Demoing lab practical work on ion channel transport in membranous systems

Feb 2017 - Jun 2017 Graduate Student Instructor for *in silico* Neuroscience

- Guiding discussions and exercises

Sep 2016 - Dec 2016 Graduate Student Instructor for Neuroscience: Cellular Mechanisms

- Reviewing exercise sets and solutions
- Guiding discussions and exercises

Aug 2010 - Dec 2010 Undergraduate Student Instructor for Anatomy Laboratory

- Demoing wet specimen and cadavers

Academic Committees

Dec 2018 - May 2019 Pint of Science, Lausanne, lead event coordinator

May 2018 - Feb 2019 ySSN young Swiss Society for Neuroscience, President

Mar 2017 - May 2018 ySSN young Swiss Society for Neuroscience, council member

Organization of Conferences

Jan 2019 1st Swiss Early-Career Researchers Symposium
Neural Population Coding and Emergent Properties of Cell Assemblies
Organizer as President of the ySSN
Campus Biotech, Geneva, Switzerland

Academic Fellowships and Awards

Sep 2022 BRAIN Initiative Individual Postdoctoral Fellowship (F32)
Postdoctoral Fellowship at Columbia University

Mar 2021 Swiss National Science Foundation Early Postdoc.Mobility Fellowship
Postdoctoral Fellowship at Columbia University

Dec 2018 EPFL-Stanford Exchange Program Fellowship
Grant of full expenses funded by Firmenich, Huguenard Lab, Stanford

Dec 2017 Exposure Science Film Hackathon
Jury voted best film, Lausanne, Switzerland

May 2017 Allen Institute Summer Workshop Selected Participant
Grant of full expenses, Seattle and Friday Harbor Laboratories, USA

Invited Presentations

Sep 2018 Blue Brain Project Review Speaker
 "The building blocks of the thalamus"
 Campus Biotech, Geneva, Switzerland

Aug 2018

Future Up! Symposium Speaker

“The somatosensory thalamus receives feedforward inhibition from local interneurons”

EPFL, Lausanne, Switzerland

Publications in peer-reviewed journals

May 2026

Local thalamic interneurons control thalamic rhythm generation and sleep-dependent learning. *Brain*. Under review.

Jane Simko*, Tamina Keira Ramirez, Caryn Martin, Sally Leung, Eleni A. Sung, Christopher D. Makinson

Sep 2025

Parallel independent voltage computing along dendrites of CA3 pyramidal neurons. *Science*. Under review.

Asako Noguchi*, Satoshi Terada, George N. Zakka, Clíodhna O’Toole, Luke Reynolds, Michelle Ann Land, **Jane Simko**, Christopher D. Makinson, Balázs J. Rózsa, François St-Pierre, Attila Losonczy

April 2025

Structural and Functional Dysregulation of Synapses Underlie Seizure and Behavioral Phenotypes in a Mouse Model

of GRIN2D Developmental and Epileptic Encephalopathy. *Brain*.

<https://doi.org/10.1093/brain/awaf125>

JiaJie Teoh*, **Jane Simko**, Chad R. Camp, Christine Liu, Wanqi Wang, Damian Williams, Liang Ma, Divyalakshmi Soundararajan, Caryn Martin, Noah K. Taylor, Ekniel François, Sabrina Petri, Ayla Kanber, Aishwarya Ravichandra, Maria Elena Pero, Francesca Bartolini, Theresa C. Swayne, Cathleen M. Lutz, Aamir Zuberi, Moran Rubinstein, Moran Hausman Kedem, Mu Yang, Hongjie Yuan, Jennifer N. Gelinis, Tristan T. Sands, Scott Q. Harper, Stephen F. Traynelis, Christopher D. Makinson, Wayne N. Frankel

July 2024

Functional architecture of intracellular oscillations in hippocampal dendrites. *Nature Comm*.

<https://doi.org/10.1038/s41467-024-50546-z>

Zhenrui Liao*, Kevin Gonzalez, Deborah Li, Catalina Yang, Donald Holder, Natalie McClain, Guofeng Zhang, Stephen Evans, Mariya Chavarha, **Jane Simko**, Christopher Makinson, Michael Lin, Adrian Negrean

Jan 2024

Behavioral control through the direct, focal silencing of neuronal activity. *Cell Chemical Biology*.

<https://doi.org/10.1016/j.chembiol.2024.04.003>

Anna V. Elleman*, Nikola Milicic, Damian J. Williams, **Jane Simko**, Christine J. Liu, Allison L. Haynes, David E. Ehrlich, Christopher D. Makinson, J. Du Bois

Mar 2023

Thalamic control of sensory processing and spindles in a biophysical somatosensory thalamoreticular circuit model of wakefulness and sleep. *Cell Reports*.

<https://doi.org/10.1016/j.celrep.2023.112200>

Elisabetta Iavarone*, **Jane Simko**, Ying Shi, Marine Bertschy, María García-Amado, Polina Litvak, Anna-Kristin Kaufmann, Christian O’Reilly, Oren Amsalem, Marwan Abdellah, Grigori Chevtchenko, Benoît Coste, Jean-Denis Courcol, Andrés Ecker, Cyrille Favreau,

Adrien Christian Fleury, Werner Van Geit, Michael Gevaert, Nadir Román Guerrero, Joni Herttuainen, Genrich Ivaska, Samuel Kerrien, James G. King, Pramod Kumbhar, Patrycja Lurie, Ioannis Magkanaris, Vignayanandam Ravindernath Muddapu, Jayakrishnan Nair, Fernando L. Pereira, Rodrigo Perin, Fabien Petitjean, Rajnish Ranjan, Michael Reimann, Liviu Soltuzu, Mohameth François Sy, M. Anil Tuncel, Alexander Ulbrich, Matthias Wolf, Francisco Clascá, Henry Markram, Sean L. Hill

- Aug 2022 Control of non-REM sleep by ventrolateral medulla glutamatergic neurons projecting to the preoptic area. *Nature Communications*.
<https://doi.org/10.1038/s41467-022-32461-3>
 Sasa Teng*, Fenghua Zhen, Li Wang, Jose Canovas Schalchli, **Jane Simko**, Xinyue Chen, Hao Jin, Christopher D. Makinson, Yueqing Peng
- Sep 2021 Morphology, physiology, and synaptic connectivity of local interneurons in the mouse somatosensory thalamus. *Journal of Physiology*.
<https://doi.org/10.1113/JP281711>
Jane Simko*, Henry Markram
- July 2021 Rodent somatosensory thalamocortical circuitry: neurons, synapses, and connectivity Review. *Neuroscience and Biobehavioral Reviews*.
<https://doi.org/10.1016/j.neubiorev.2021.03.015>
 Christian O'Reilly*, Elisabetta Iavarone*, **Jane Yi***, and Sean Hill
- May 2019 Experimentally-constrained biophysical models of tonic and burst firing modes in thalamo-cortical neurons. *PLOS Computational Biology*.
<https://doi.org/10.1371/journal.pcbi.1006753>
 Elisabetta Iavarone*, **Jane Yi**, Ying Shi, Bas-Jan Zandt, Christian O'Reilly, Werner Van Geit, Christian Rössert, Henry Markram, and Sean Hill

*indicates first-authorships

Contributions to International Conferences

- Oct 2022 Functional interrogation of the mouse somatosensory thalamic interneuron in sensory perception and rhythmic states
Jane Simko, Chris Makinson
 Gordon Research Conference, Thalamocortical Interactions, poster presentation
 Lucca, Italy
- Oct 2019 Detailed computational model of thalamic microcircuitry
 Elisabetta Iavarone, **Jane Yi**, Ying Shi, Henry Markram, Sean Hill
 SfN, Society for Neuroscience, poster presentation
 Chicago, IL
- Nov 2018 The building blocks of first order thalamic nuclei
Jane Yi, Rodrigo Perin, Ying Shi and Henry Markram
 SfN, Society for Neuroscience, poster presentation
 San Diego, CA
- Feb 2018 The inhibitory influences in the somatosensory thalamus

Jane Yi, Rodrigo Perin and Henry Markram
GRC, Gordon Research Conference, Thalamocortical interactions,
poster presentation
Lucca, Italy

July 2018 The somatosensory thalamus receives feedforward inhibition from local interneurons

Jane Yi, Elisabetta Iavarone, Rodrigo Perin, Ying Shi and Henry Markram
FENS, Federation of European Neuroscience Societies, poster presentation
Berlin, Germany

Jul 2018 Data-driven models of interneurons in the somatosensory thalamus and comparison with gene expression data
Elisabetta Iavarone, Jane Yi, Ying Shi, Christian O'Reilly, Werner Alfons Hilda van Geit, Christian A Rössert, Henry Markram, Sean Hill
OCNS, Organization for Computational Neurosciences, poster presentation
Seattle, WA

Nov 2017 The inhibitory influences in the somatosensory thalamus
Jane Yi, Elisabetta Iavarone, Christian O'Reilly, Ying Shi, Rodrigo Perin and Henry Markram
SfN, Society for Neuroscience, poster presentation
Washington, D.C

Nov 2016 A somatotopic distribution of physiological, morphological, and connectivity properties in the ventral posterolateral nucleus
Jane Yi, Elisabetta Iavarone, Christian O'Reilly, Rodrigo Perin and Henry Markram
SfN, Society for Neuroscience, poster presentation
San Diego, CA

Nov 2016 A data-driven computational model of relay cells of the ventral posterolateral nucleus of the thalamus
Elisabetta Iavarone, Christian O'Reilly, Jane Yi, Sean Hill, and Henry Markram
SfN, Society for Neuroscience, poster presentation
San Diego, CA

Scientific Outreach Activities and Events

Jan 2019 1st Swiss Early-Career Researchers Symposium
Neural Population Coding and Emergent Properties of Cell Assemblies
Organizer as President of the ySSN
Campus Biotech, Geneva, Switzerland

May 2019 Pint of Science
Global Science Festival, lead event coordinator
Le Lab du Dr. Gab's, Lausanne, Switzerland

Dec 2017 Exposure Science Film Hackathon
Hackathon participant for science communication, Film awardee
Lausanne, Switzerland

Publicly Available Resources

- Sep 2025 Local thalamic interneurons drive spindle termination and enable sleep-dependent learning. *BioRxiv*.
<https://doi.org/10.1101/2025.09.26.678870>
Jane Simko*, Tamina Keira Ramirez, Caryn Martin, Sally Leung, Eleni A. Sung, Christopher D. Makinson
- Feb 2024 Functional architecture of intracellular oscillations in hippocampal dendrites. *BioRxiv*.
<https://doi.org/10.1101/2024.02.12.579750>
Zhenrui Liao*, Kevin Gonzalez, Deborah Li, Catalina Yang, Donald Holder, Natalie McClain, Guofeng Zhang, Stephen Evans, Mariya Chavarha, **Jane Yi**, Christopher Makinson, Michael Lin , Adrian Negrean
- Mar 2022 Thalamic control of sensory enhancement and sleep spindle properties in a biophysical model of thalamoreticular microcircuitry. *BioRxiv*.
<https://doi.org/10.1101/2022.02.28.482273>
Elisabetta Iavarone*, **Jane Simko**, Ying Shi et al.
- July 2020 Rodent somatosensory thalamocortical circuitry: neurons, synapses, and connectivity. Review. *Preprints*.
<https://doi.org/10.20944/preprints202007.0053.v1>
Christian O'Reilly*, Elisabetta Iavarone*, **Jane Yi***, and Sean Hill
- May 2019 Thalamocortical relay models constrained by experiment and optimization.
ModelDB, accession number: 251881
Elisabetta Iavarone, **Jane Yi**, Ying Shi, Bas-Jan Zandt, Christian O'Reilly, Werner Van Geit, Christian Rössert, Henry Markram, and Sean Hill
- Jan 2019 Experimentally-constrained biophysical models of tonic and burst firing modes in thalamo-cortical neurons. *BioRxiv*.
<https://doi.org/10.1101/512269>
Elisabetta Iavarone*, **Jane Yi**, Ying Shi, Bas-Jan Zandt, Christian O'Reilly, Werner Van Geit, Christian Rössert, Henry Markram, and Sean Hill

**indicates first-authorships*

Skills

- Programming* Python
C++
MATLAB
- Languages* English native fluency
French proficient fluency
Korean proficient fluency

Miscellaneous

- Violin, 22 years of study
Ballet, 17 years of study